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Course Code: INT102R01
Semester: IV

FUNDAMENTALS OF DATABASE MANAGEMENT SYSTEMS

Course Objectives:

This course will help the learner to infer the fundamentals of data models and to conceptualize and depict a database system using ER diagram. Construct tables and write effective queries to design and implement a database for real-world applications.

UNIT - I

11 Periods

Introduction: Data-Database - Characteristics of the Database Approach - Actors on the scene - Advantages - Data models, Schemas and Instances - Three Schema Architecture and Data Independence - Database Languages and Interfaces - Database System Environment. **Data Modeling:** Data modeling using the ER model

UNIT - II

11 Periods

SQL: Basic SQL - Data Definition and Data Types, Specifying Constraints, Basic Retrieval Queries, Insert, Delete and Update statements - Additional features of SQL - More Complex SQL Retrieval Queries - Specifying Constraints as Assertions and Actions as Triggers - Views - Schema Change Statements in SQL - JDBC - SQL Class Library for Java Programming

UNIT - III

12 Periods

Relational Database Design: Functional Dependency - First, Second and Third Normal Forms - BCNF - Multi valued Dependency and Fourth Normal Form - Join Dependency and Fifth Normal Form. **Transaction Processing Concepts:** Introduction to Transaction Processing - Transaction and System Concepts - Serializability - Concurrency Control Techniques - Two-Phase Locking Techniques, Timestamp Ordering

UNIT - IV

11 Periods

Introduction to Non-Relational Databases (MongoDB): Getting Started - Documents, collections, databases, data types. **Creating, Updating and Deleting documents:** Inserting and saving documents - removing documents - updating documents - Querying - Introduction to find querying - Type-specific Queries

TEXT BOOKS

1. R.Elmasri, S.B.Navathe. *Fundamentals of Database Systems*, Addison Wesley, Seventh Edition, 2016.
2. Kristina Chodorow and Michel Dirolf. *MongoDB: The Definitive Guide*, O'Reilly, Publications, Second Edition, 2013.

REFERENCES

1. Henry F.Kort, Abraham Silberschatz, Sudarshan. *Database System Concepts*, McGraw Hill, Seventh Edition, 2019.
2. S.K.Singh. *Database Systems Concepts, Design and Applications*, Prentice Hall, First Edition, 2009.
3. Raghu Ramakrishnan, Johannes Gehrke. *Database Management Systems*, McGraw Hill, Third Edition, 2014.

ONLINE MATERIALS

1. <https://nptel.ac.in/courses/106105175/1>
2. <https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-830-database-systems-fall-2010/lecture-notes/>

UNITWISE LEARNING OUTCOMES

Upon successful completion of each unit, the learner will be able to

Unit I	• Compare and contrast database systems from file systems and describe the fundamental elements of relational database management systems • Design ER-models to represent simple database application scenarios
Unit II	• Develop SQL queries for the given scenario using DDL and DML commands • Construct connectivity with database using JDBC
Unit III	• Improve the database design by normalization • Design and implement various concurrency control techniques in transaction processing system
Unit IV	• Analyze the recent advancements in MongoDB • Develop and run the application using MongoDB

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to

- Illustrate the fundamental elements of relational database management systems
- Design ER,EER-models to represent simple database application scenarios
- Experiment with SQL to construct queries for the given problem
- Improve the database design using normalization techniques
- Design and implement non relational databases for real world applications
- Develop and run the application using MongoDB